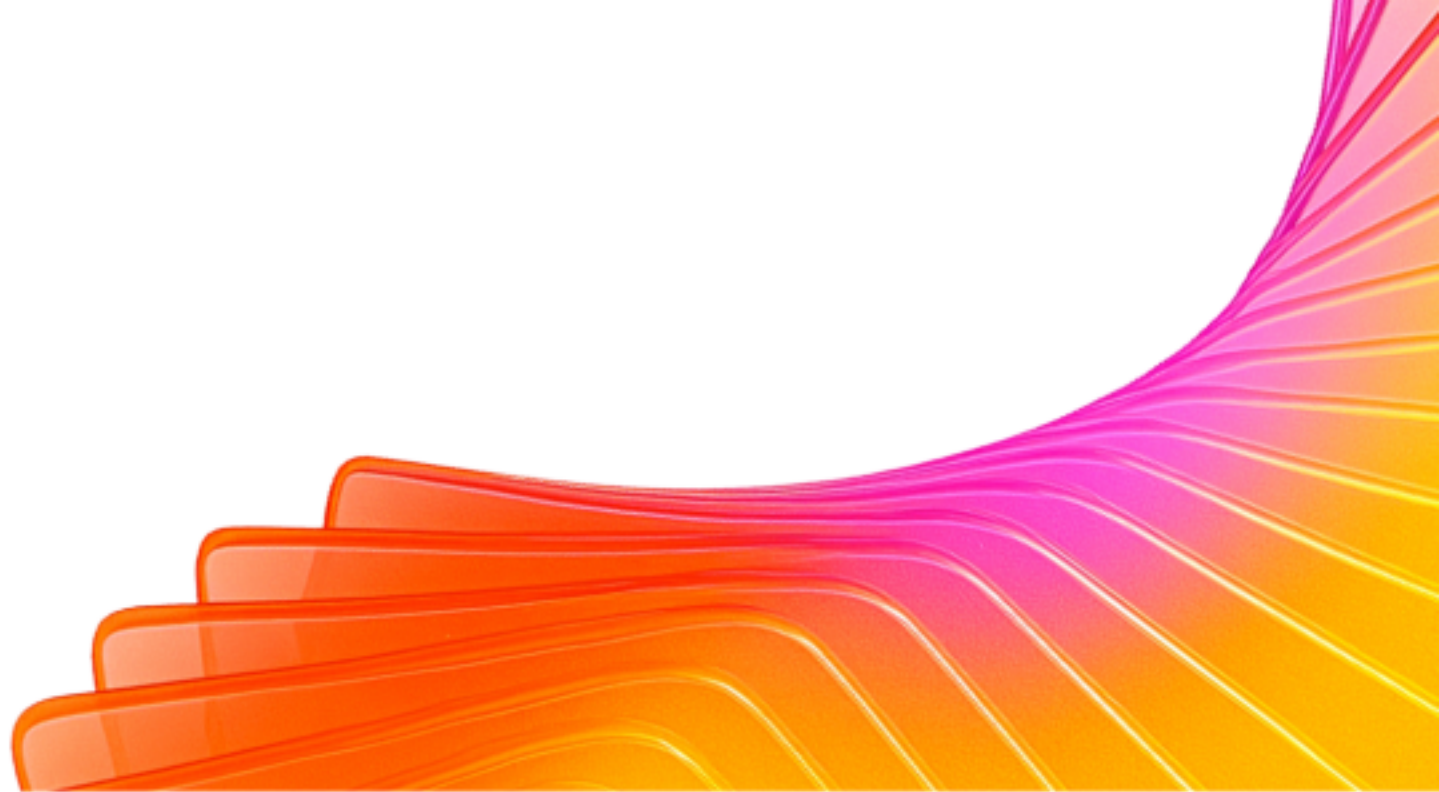


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Participant Details

Participant Name: Anuranjan Jain

Team Name: Wizard

Team Role: Solo participant and team leader

Brief about the idea

CivicMind AI is an AI-powered Decision Intelligence Platform for smart city operations. It combines civic datasets, BigQuery analytics, Gemini-powered agents, RAG over city SOPs, and Looker dashboards to help public teams detect issues, ask questions in natural language, forecast risk, and choose the next best action.

The prototype focuses on urban operations: citizen complaints, waste collection, waterlogging, utility outages, traffic incidents, environmental metrics, and public-service workflows. Instead of showing disconnected reports, it turns scattered community data into explainable recommendations that city teams can review and act on.

How the solution meets the challenge

Problem fit

Fragmented civic data slows decisions across complaints, transport, utilities, waste, environment, and emergency workflows.

Google Cloud build

Storage and Functions ingest data; BigQuery analyzes it; Gemini and ADK coordinate multi-agent reasoning.

Decision output

RAG grounds answers in SOPs; agents rank actions; Cloud Run and Looker deliver app, dashboard, and reports.

Real-world impact: city teams detect spikes earlier, prioritize scarce resources, explain decisions with evidence, and keep humans in the approval loop before any civic action is executed.

Core workflow: ask a question -> retrieve trusted data -> detect anomalies -> apply policy context -> recommend next action -> generate an approval-ready summary.

Opportunities

Different from existing ideas
Not just a chatbot or dashboard.
CivicMind combines analytics, forecasting, RAG, recommendations, and operational workflow in one platform.

USP
Every action is explainable: tied to BigQuery evidence, anomaly signals, policy/SOP grounding, owner, severity, and expected impact.

Scale advantage
The same modular Cloud Run + BigQuery + ADK architecture can serve a campus, ward, district, or full smart-city deployment.

Human-in-the-loop by design: AI recommends and explains; city officials approve and act. That makes the solution practical for sensitive civic operations where accountability matters.

Unique pitch: a civic operations co-pilot that turns noisy community signals into ranked, auditable interventions.

List of features offered by the solution

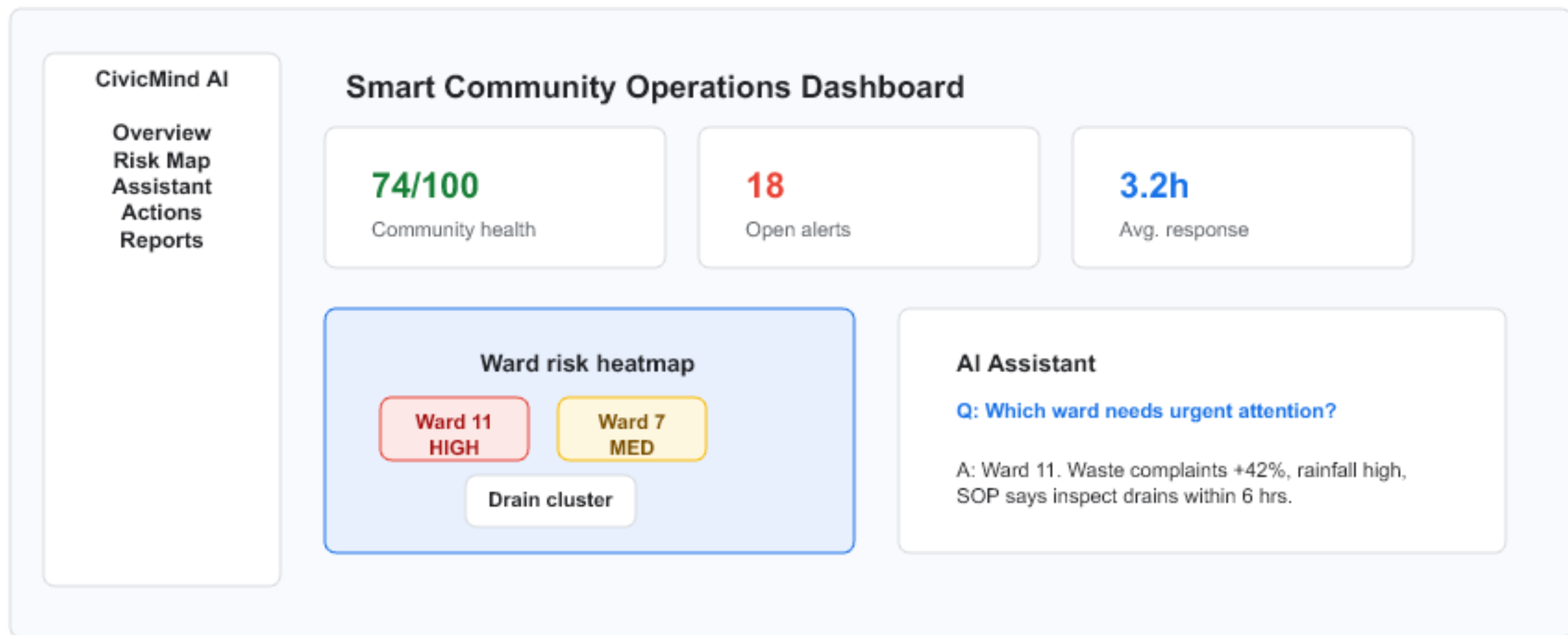
1. Natural-language city analytics over BigQuery civic datasets.
2. Ward-level risk scoring for complaints, traffic, utilities, waste, and environment.
3. Anomaly detection for sudden spikes such as garbage overflow, outage clusters, waterlogging, or congestion.
4. AI-generated action recommendations with severity, affected population, owner, confidence, and expected impact.
5. RAG-backed policy assistant for SOPs, escalation rules, disaster guides, and civic service documents.
6. Executive Looker dashboard for KPIs, heatmaps, category trends, open alerts, and response status.
7. Daily/weekly Gemini reports for ward officers and administrators.
8. Responsible-AI controls: citations, uncertainty, PII minimization, audit logging, and human approval.

Process flow diagram or Use-case diagram



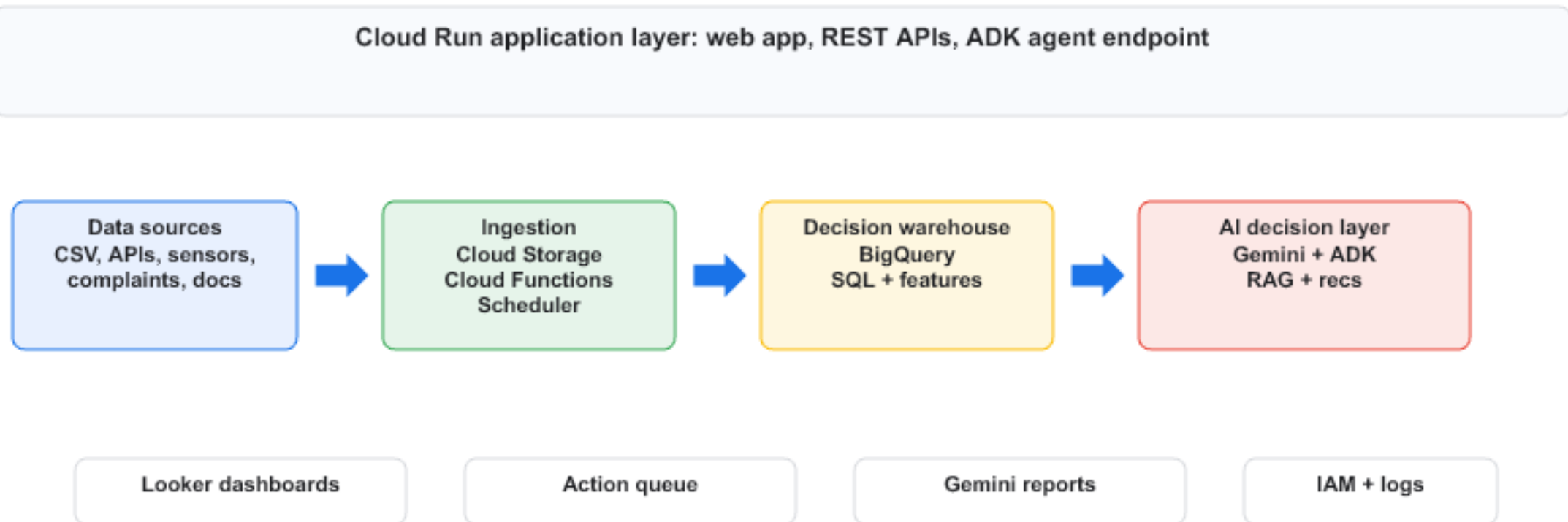
Use case: "Which ward needs urgent attention today?" CivicMind retrieves data, detects abnormal patterns, applies SOP context, and produces an approval-ready decision brief.

Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:

Scalable deployment architecture for a ward, campus, district, or full-city rollout



Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Google Cloud services used

- BigQuery: analytics warehouse, SQL insights, ward-level risk features, natural-language data grounding.
- Cloud Storage: raw data lake for CSVs, civic docs, images, and historical reports.
- Cloud Functions / Cloud Run jobs: event-driven ingestion, cleaning, scheduled refresh, and alert generation.
- Cloud Run: scalable backend APIs, ADK agent service, and web prototype deployment.
- Gemini Enterprise Agent Platform / Vertex AI + Gemini: reasoning, summarization, recommendations, and report generation.
- Agent Development Kit: coordinator, analyst, anomaly, recommendation, policy/RAG, and report agents.
- Vector Search / RAG: trusted retrieval from SOPs, policies, emergency guides, and historical reports.
- Looker / Looker Studio: dashboards, heatmaps, category trends, and stakeholder reporting.
- Cloud Scheduler, Logging, Monitoring, IAM, Secret Manager: production-readiness, reliability, and secure operations.

Snapshots of the prototype

Dashboard snapshot

High

Risk score

Complaint trend
Waste +42% | Water +31%

AI assistant snapshot

Q: Which ward needs urgent attention today?

A: Ward 11. Waste complaints rose 42%, rainfall is above threshold, and drain inspection SOP applies. Recommend extra pickup, drain inspection, and citizen advisory.

Recommendation queue

1. Dispatch waste team
Owner: Public Works | 6 hrs
2. Inspect drains
Owner: Ward Engineer | 8 hrs
3. Issue citizen advisory
Owner: Comms | 2 hrs

Google Cloud 

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Thank you

